

MTH2021 Mathematical Methods 2

Homework set 1

1. Let $z_1 = 3 - 2i$ and $z_2 = -1 + i$. Write the following expressions in the Cartesian form $z = x + iy$:

(a) $z_1 - 2z_2$

(b) $z_1 z_2$

(c) z_1/z_2

(d) $z_1 + z_2^*$

2. Express the results of Question 1 in polar form $z = r \exp(i\theta)$, where $r = |z|$ and $\theta = \text{Arg } z$ are the modulus and the principal value of the argument of z .

3. On separate Argand diagrams, sketch the lines/curves/regions for z represented by the equations/inequalities

(a) $z = 1 + it$ for $0 \leq t \leq 2$

(b) $z = t + it^2$ for $-1 < t < 1$

(c) $z = \cos t + i \sin t$ for $0 < t \leq 2\pi$

(d) $|z - 2i| = 2$

(e) $|z + 1| \leq 1$

4. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, show that $(z_1 z_2)^* = z_1^* z_2^*$.

5. Prove De Moivre's theorem by mathematical induction.

6. Use De Moivre's theorem to evaluate

(a) $(1 + i)^8$

(b) $(1 + i)^8 - (1 - i)^8$

(c) $\left(\frac{1 + i}{1 + i\sqrt{3}} \right)^{12}$

7. Use De Moivre's theorem to show that

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$$

8. Find all distinct solutions of the equations

(a) $2z^2 + 2z + 1 = 0$

(b) $z^4 = -1$

(c) $z^3 = 1 + i$

(d) $z^3 = -8$
